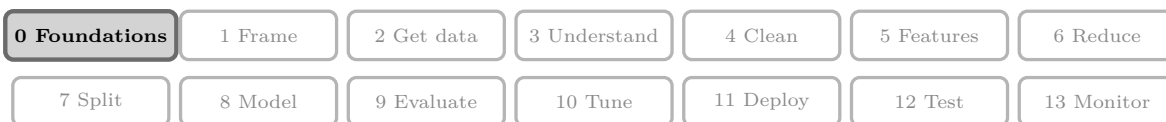


Convex Sets, Convex Functions and Convex Optimisation

Contents

1. Overview	2
2. Convex sets	3
2.1 The definition	3
2.2 Combining convex sets	4
3. Convex functions and their sets	5
4. Testing convexity with derivatives	6
4.1 Curving up: the slope keeps increasing	6
4.2 First-order condition: tangents lie below	8
4.3 Second-order condition in several variables	9
5. Building convex functions	10
6. Convex optimisation problems	12
6.1 The definition	12
6.2 What convexity guarantees	12
6.3 Which ML problems are convex	13
7. Summary	15
8. Sources	15
9. Key terms	15

Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).

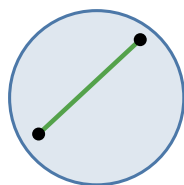


- **Builds on:** Convex and non-convex loss ([Note ML-056](#)); Lagrange multipliers, KKT and duality ([Note MA-066](#)).
- **Leads to:** Linear and quadratic programming ([Note MA-068](#)); Local minima and saddle points ([Note DL-032](#)).

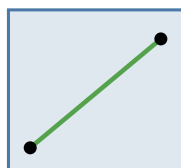
1. Overview

Key point: A set is convex when the segment between any two of its points stays inside it; a convex optimisation problem minimises a convex function over a convex set, and for such problems every local minimum is global and the dual gives the same answer as the primal.

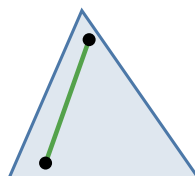
convex



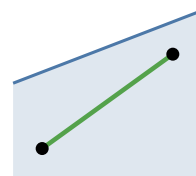
disc



box

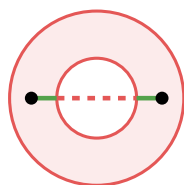


triangle

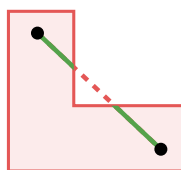


half-plane (part)

non-convex



ring



L-shape



star

dashed red: the
part of the segment
outside the set

Figure 1: Convex sets (top): every segment between two points stays inside. Non-convex sets (bottom): some segment leaves the set

Two Notes prepare this one. The [convex and non-convex cost functions Note](#) defined a convex function by the chord test and showed why gradient descent likes them. The [Lagrange multipliers Note](#) added constraints, which cut the parameter space down to a feasible region. This Note puts the two together. Figure 1 shows the new ingredient, convex sets. Then come:

- convex functions and their link to convex sets (Section 3);
- three practical tests for convex functions: a slope that keeps increasing, tangent lines below the graph, and the Hessian (Section 4);
- rules for building convex functions (Section 5);
- the convex optimisation problem (Section 6).

2. Convex sets

Key point: A set C is convex if, for any two points in it, every point on the segment between them is also in C .

2.1 The definition

Key point: The segment between $\mathbf{x}$ and $\mathbf{y}$ is all points $\theta\mathbf{x} + (1 - \theta)\mathbf{y}$ with $0 \leq \theta \leq 1$; all of them must lie in the set.

The points between $\mathbf{x}$ and $\mathbf{y}$ are written the same way as in the chord test of the [convex and non-convex cost functions](#) **Note:** a mix $\theta\mathbf{x} + (1 - \theta)\mathbf{y}$, with θ from 0 to 1.

1. **In words:** pick any two points of the set; the whole straight segment joining them must stay in the set.
2. **Formula:** C is a **convex set** (G-479) if for all $\mathbf{x}, \mathbf{y} \in C$ and all θ with $0 \leq \theta \leq 1$,

$$\theta\mathbf{x} + (1 - \theta)\mathbf{y} \in C$$

3. **Example:** the disc of radius 1, all points with length at most 1. Take $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (0, 1)$, both on its edge. The midpoint ($\theta = 0.5$) is $(0.5, 0.5)$, with length $\sqrt{0.25 + 0.25} = 0.71 \leq 1$: inside.

Now the ring of points with length between 1 and 2. Take $\mathbf{x} = (1.5, 0)$ and $\mathbf{y} = (-1.5, 0)$, both in the ring. Their midpoint is $(0, 0)$, with length $0 < 1$: outside. One such pair is enough, so the ring is not convex.

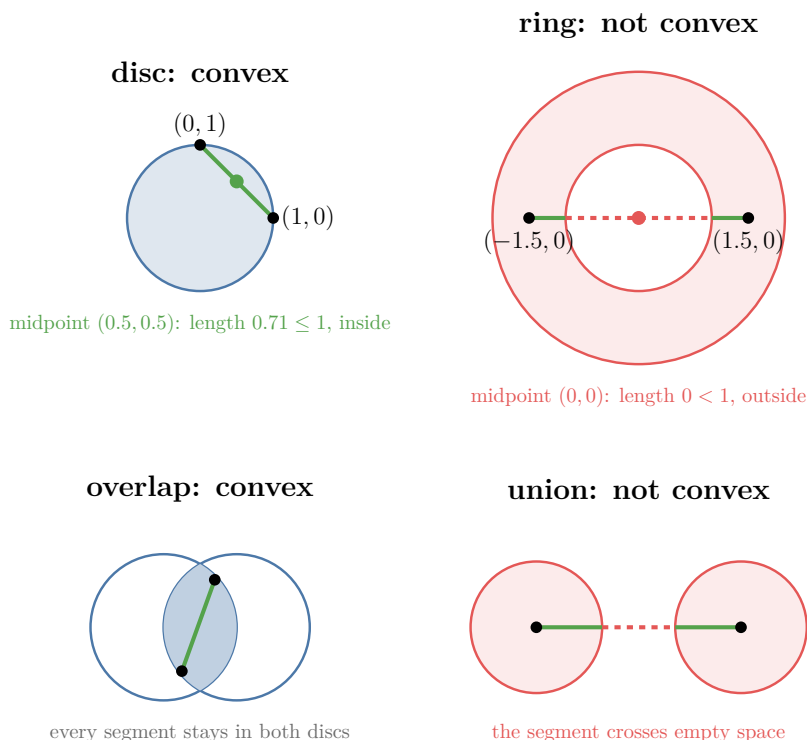


Figure 2: Top: the disc passes the midpoint test, the ring fails it at $(0, 0)$. Bottom: the overlap of two discs is convex; the union of two separate discs is not

Figure 2 draws both examples and the two ways of combining sets from Section 2.2. Watch the red dashed part: one segment leaving the set is enough to make it non-convex.

Informally, a convex set has no dents and no holes (Figure 1). Common convex sets in ML:

- **a line, a plane or a hyperplane** (see the [equation of a hyperplane Note](#)), the set where a linear equation holds;
- **a half-space**, one side of a hyperplane, such as $x + y \geq 3$ in the [Lagrange multipliers Note](#);
- **a box**, such as $-1 \leq x_i \leq 1$ for every coordinate;
- **a ball** $\|\mathbf{w}\| \leq t$, the Ridge constraint, and the diamond $|w_1| + |w_2| \leq t$, the Lasso constraint.

2.2 Combining convex sets

Key point: The overlap (intersection) of convex sets is convex; their union usually is not.

If $\mathbf{x}$ and $\mathbf{y}$ lie in both A and B , the segment between them lies in A (because A is convex) and in B (because B is convex), so it lies in the overlap. The overlap rule is why a feasible region made of many convex constraints is convex: it is the overlap of one convex set per constraint. A box, for example, is the overlap of $2n$ half-spaces.

A union does not keep convexity. Two separate discs together form a set where the segment from one disc to the other crosses empty space.

Extra: The set where a convex function stays at or below a level, $\{\mathbf{x} : g(\mathbf{x}) \leq c\}$, is always convex (Boyd and Vandenberghe §3.1.6). The level-set rule is why convex optimisation asks for convex functions g_i in its inequality constraints $g_i(\mathbf{x}) \leq 0$ (Section 6): each constraint then cuts out a convex set. The set where a convex function is exactly 0 is not convex in general (the circle $x^2 + y^2 = 1$ is not), which is why equality constraints must be linear.

3. Convex functions and their sets

Key point: A function is convex when every chord lies on or above its graph; equivalently, the region above its graph is a convex set.

The [convex and non-convex cost functions](#) Note defined a **convex function**: $f(\theta\mathbf{a} + (1-\theta)\mathbf{b}) \leq \theta f(\mathbf{a}) + (1-\theta)f(\mathbf{b})$ for all $\mathbf{a}$, $\mathbf{b}$ and $0 \leq \theta \leq 1$. One condition was left implicit there: the domain of f must itself be a convex set, so that the mixed point $\theta\mathbf{a} + (1-\theta)\mathbf{b}$ is somewhere f is defined.

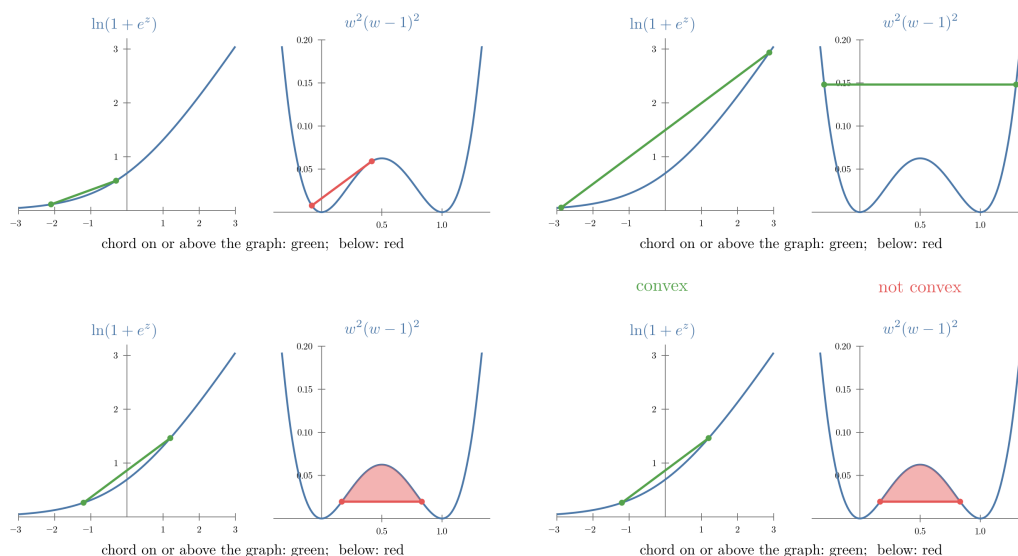


Figure 3: The chord test. A chord joins two points of each graph and slides. Left: softplus, every chord stays on or above the graph (green). Right: $w^2(w - 1)^2$, the chord from $w = 0.17$ to 0.83 runs below the hump (red gap), so the function is not convex

Figure 3 runs the definition. Watch for red: one chord below the graph is enough to break convexity, while a convex function never shows red, wherever the two ends are placed. A **strictly convex** function, whose curve lies strictly below every chord, has at most one minimum, which is why a strictly convex loss leaves gradient descent nowhere to get stuck.

Two related ideas:

- **Concave function:** (G-438) the negative of a convex function, an upside-down bowl. Every chord lies on or below the graph. The natural logarithm is concave; so is the dual function $D(\lambda) = 3\lambda - 3\lambda^2/8$ of the [Lagrange multipliers Note](#). Maximising a concave function is the same task as minimising a convex one.
- **Epigraph:** (G-695) the region on and above the graph of f , as if the bowl were filled with water. A function is convex exactly when its epigraph is a convex set (Boyd and Vandenberghe §3.1.7). This links the two halves of this Note: convex functions are convex sets seen from above.

Extra: The defining inequality is the two-point case of **Jensen's inequality** (G-982). With more points and weights $\theta_i \geq 0$ that add up to 1, a convex f satisfies $f(\sum_i \theta_i \mathbf{x}_i) \leq \sum_i \theta_i f(\mathbf{x}_i)$: the function of an average is at most the average of the function. For $f(x) = x^2$ and the points 0, 1, 2 with equal weights: $f(1) = 1 \leq (0 + 1 + 4)/3 = 1.67$. With probabilities as weights, this reads $f(E[X]) \leq E[f(X)]$ (see the [expected value and variance Note](#)); for $f(x) = x^2$ it says $E[X^2] - (E[X])^2 \geq 0$, that is, a variance is never negative.

4. Testing convexity with derivatives

Key point: In one variable, a convex function is one whose slope never decreases, so $f''(x) \geq 0$; equivalently every tangent lies on or below the graph. In several variables, the Hessian must have no negative eigenvalues anywhere.

Checking every chord is impossible in practice. When derivatives exist, three shortcuts replace it.

4.1 Curving up: the slope keeps increasing

Key point: Where a curve bends upwards like a cup, its slope is increasing, so its second derivative is ≥ 0 . A convex function bends upwards everywhere.

Walk along a curve from left to right and watch its slope.

1. **In words:** on a cup-shaped part, the slope starts negative, rises through 0 at the bottom, and keeps rising. On a hump-shaped part, the slope does the opposite: it falls from positive through 0 to negative. A rising slope means the derivative of the slope, the second derivative, is positive.
2. **The terms:** calculus books call a cup-shaped part **concave up** and a hump-shaped part **concave down** (G-2256). Concave up is the same as convex, and concave down is the same as concave. The two names describe one shape, so they never disagree: the chord test of Section 3 and the rising slope of this section pick out exactly the same functions, for any function with a second derivative (Boyd and Vandenberghe §3.1.4).
3. **Example:** the function $q(w) = w^2(w - 1)^2$ of Figure 3, with slope $q'(w) = 2w(w - 1)(2w - 1)$ and second derivative $q''(w) = 12w^2 - 12w + 2$.

- q'' is 0 at $w = 0.211$ and $w = 0.789$, the **inflection points** (G-945) where the bending changes direction.
- Outside them $q'' > 0$: q curves up, convex there. Between them $q'' < 0$: at $w = 0.5$, $q''(0.5) = 3 - 6 + 2 = -1$, so q curves down there.
- One stretch with $q'' < 0$ is enough: q is not convex, as the chord test found.

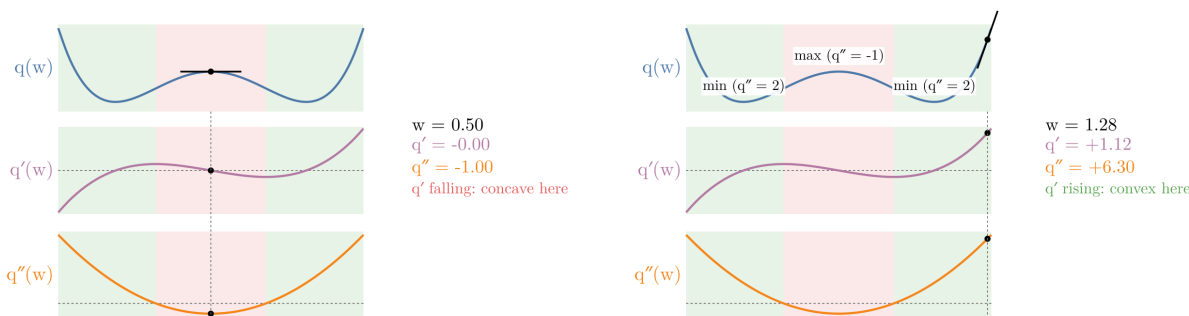


Figure 4: Three stacked graphs share one moving cursor: $q(w) = w^2(w-1)^2$ with its tangent line, its slope $q'(w)$, and its second derivative $q''(w)$. Green where $q'' \geq 0$ (the slope rises, q curves up), red where $q'' < 0$. At the end the three flat points are sorted by the second derivative test. Idea after Khan Academy, “Concavity introduction”

In Figure 4, watch the tangent line and the middle graph together. In the green parts the tangent turns anticlockwise and q' climbs; in the red part the tangent turns clockwise and q' falls.

The second derivative test. At a point c where the tangent is flat, $f'(c) = 0$:

- $f''(c) > 0$: the curve is a cup there, so c is a local minimum;
- $f''(c) < 0$: the curve is a hump, so c is a local maximum;
- $f''(c) = 0$: the test says nothing.

For q : $q'(0) = 0$ and $q''(0) = 2$, a minimum; $q'(0.5) = 0$ and $q''(0.5) = -1$, a maximum; $q'(1) = 0$ and $q''(1) = 2$, a minimum. This is the **second derivative test** (G-2257).

The one-variable condition. A function with a second derivative is convex exactly when $f''(x) \geq 0$ for every x : it curves up, or stays straight, everywhere. Then every flat point passes the second derivative test as a minimum, never a maximum.

- **Example:** the **softplus** (G-1834) function $f(z) = \ln(1 + e^z)$. Its derivative is the **sigmoid** $\sigma(z)$, a slope that rises steadily from 0 to 1. Its second derivative is $f''(z) = \sigma(z)(1 - \sigma(z))$, the sigmoid’s derivative (see the **sigmoid derivative Note**). Both factors lie between 0 and 1, so $f''(z) > 0$ everywhere: convex. At $z = 0$, $f''(0) = 0.5 \times 0.5 = 0.25$.

4.2 First-order condition: tangents lie below

Key point: f is convex exactly when $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$: the tangent plane at any point underestimates f everywhere.

The right-hand side is the tangent plane of f at $\mathbf{x}$, the first-order Taylor approximation of the [Hessian and multivariate Taylor Note](#) (Section 3).

1. **In words:** the tangent line or plane at any point never rises above the graph. This test is the **first-order condition** (G-781).
2. **Formula:** a differentiable f is convex if and only if, for all $\mathbf{x}$ and $\mathbf{y}$ (Boyd and Vandenberghe §3.1.3),

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$$

3. **Example:** softplus, $f(z) = \ln(1 + e^z)$, with derivative $\sigma(z)$ (Section 4.1). At $z = 0$ the slope is $\sigma(0) = 0.5$ and the value is $\ln 2 = 0.693$. At $z = 2$:

$$f(2) = \ln(1 + e^2) = 2.13, \quad \text{tangent: } 0.693 + 0.5 \times (2 - 0) = 1.69$$

$2.13 \geq 1.69$: the curve is above its tangent (Figure 5, left).

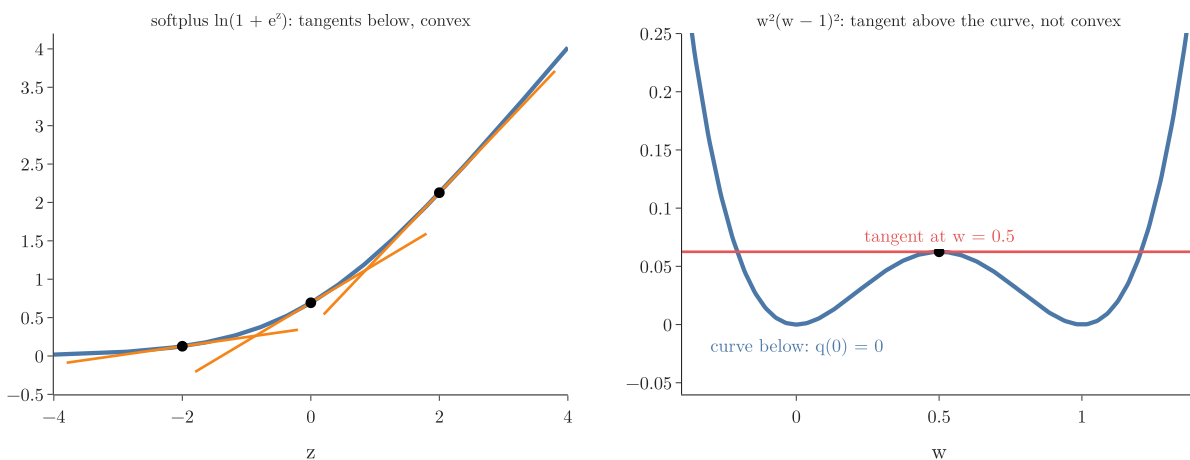


Figure 5: The first-order test. Left: every tangent of softplus lies below the curve, so it is convex. Right: the tangent of $w^2(w - 1)^2$ at $w = 0.5$ lies above the curve near 0 and 1, so it is not convex

Figure 5 (right) shows the test failing. For $q(w) = w^2(w - 1)^2$, the tangent at $w = 0.5$ is flat at height 0.0625, but $q(0) = 0$ lies below it.

The tangent test and the rising slope of Section 4.1 are one fact seen twice. If the slope never decreases, the curve to the right of a point rises at least as fast as the tangent there, and the curve to the left falls at least as fast, so the tangent can only stay below (Boyd and Vandenberghe §3.1.3).

The first-order condition gives the most useful fact about convex functions in one line. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$, the condition becomes $f(\mathbf{y}) \geq f(\mathbf{x}^*)$ for every $\mathbf{y}$. So for a convex function, any point with zero gradient is a global minimum. Gradient descent stops at zero gradient, so on a convex function it stops at the best answer.

Extra: Softplus is the [log loss](#) in disguise. For an **observation** (one record, a row of the data table) with label 0 and score z , the loss $-\ln(1 - \sigma(z))$ equals $\ln(1 + e^z)$; for label 1, $-\ln \sigma(z) = \ln(1 + e^{-z})$. Both are convex in z , and $z = \mathbf{w}^\top \mathbf{x}$ is linear in the weights. A convex function of a linear function is convex (Boyd and Vandenberghe §3.2.2), so the loss of logistic regression is convex in $\mathbf{w}$: gradient descent on it reaches the global minimum.

4.3 Second-order condition in several variables

Key point: A twice-differentiable f is convex exactly when its Hessian is positive semi-definite (no negative eigenvalues) at every point; in one variable, $f''(x) \geq 0$ everywhere.

With several variables the function must curve up along every direction, and the second derivative becomes the **Hessian matrix** (G-888). The [Hessian and multivariate Taylor Note](#) (Section 5.3) showed that the signs of the Hessian's eigenvalues give the local shape: all positive is a bowl, mixed signs a saddle. A symmetric matrix with no negative eigenvalues is **positive semi-definite** (G-1532) (see the [SVD geometry Note](#)). Convexity asks for this at every point, not just one.

1. **In words:** the function must curve upwards (or stay flat) in every direction, everywhere. This test is the **second-order condition** (G-1761).
2. **Formula:**

$$f \text{ is convex} \iff \nabla^2 f(\mathbf{x}) \text{ is positive semi-definite for every } \mathbf{x}$$

In one variable: $f''(x) \geq 0$ for every x .

3. **Example:** softplus in one variable, $f''(z) > 0$ everywhere (Section 4.1).

With two variables, compare two quadratic functions. Their Hessians are constant, so one check covers every point:

Function	Hessian	Eigenvalues	Convex?
$x^2 + xy + y^2$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	1 and 3	yes: a bowl
$x^2 + 3xy + y^2$	$\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$	5 and -1	no: a saddle

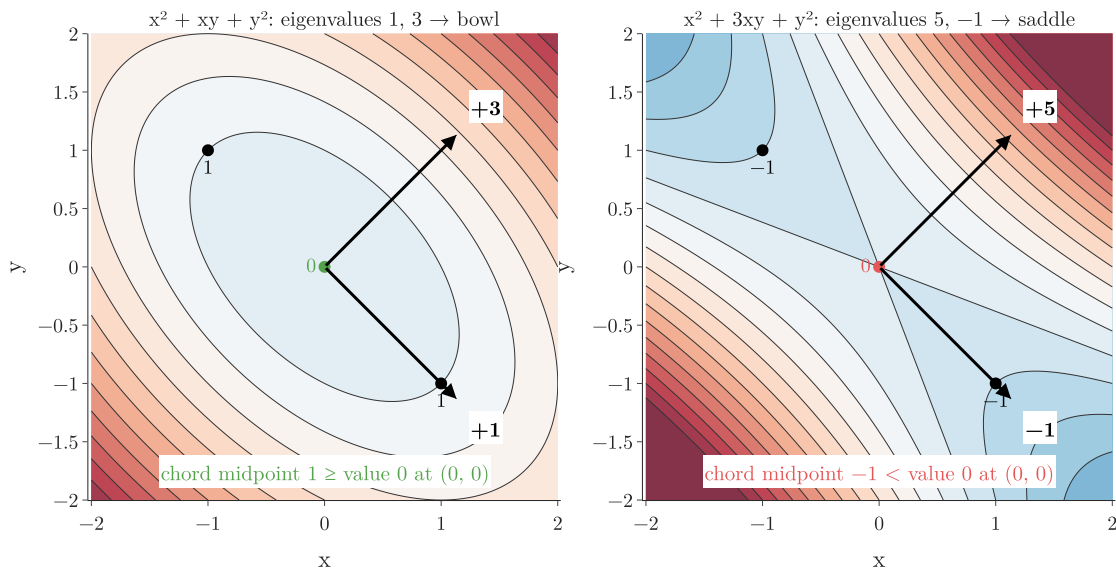


Figure 6: Contour maps of the two quadratics. Arrows: the Hessian’s eigenvector directions, labelled with their eigenvalues. Left: both positive, closed ellipses, a bowl. Right: one negative, a saddle; the chord between $(1, -1)$ and $(-1, 1)$ sits at -1 , below the value 0 at the midpoint

In Figure 6, watch the sign of each arrow’s eigenvalue: along the arrow labelled -1 , the saddle curves down, and that one direction is enough to break convexity.

The chord test agrees. For $x^2 + 3xy + y^2$, the points $(1, -1)$ and $(-1, 1)$ both give $1 - 3 + 1 = -1$. Their midpoint $(0, 0)$ gives 0 , above the chord value -1 .

Python: the eigenvalue check in NumPy. `eigvalsh` is meant for symmetric matrices such as Hessians and returns the eigenvalues in increasing order.

```
import numpy as np
```

```
H1 = np.array([[2, 1], [1, 2]])
H2 = np.array([[2, 3], [3, 2]])
np.linalg.eigvalsh(H1)    # [1. 3.] all >= 0: convex
np.linalg.eigvalsh(H2)    # [-1. 5.] one < 0: not convex
```

5. Building convex functions

Key point: Adding convex functions, or scaling them by non-negative numbers, keeps them convex; this is why “convex loss plus convex penalty” is convex.

In practice we rarely test a function from scratch. We build it from pieces known to be convex, using rules that keep convexity.

Non-negative weighted sums. If f_1 and f_2 are convex and $\alpha, \beta \geq 0$, then $\alpha f_1 + \beta f_2$ is

convex. The proof is one step: write the chord inequality for f_1 and for f_2 , multiply each by its non-negative weight (which keeps the direction of $\leq$), and add them.

Worked with numbers: $f_1(w) = w^2$ and $f_2(w) = |w|$, with $\alpha = 1$, $\beta = 2$, between $a = -1$ and $b = 3$ at the midpoint 1:

$$\text{curve: } 1^2 + 2 \times |1| = 3, \quad \text{chord: } 0.5 \times (1 + 2) + 0.5 \times (9 + 6) = 9$$

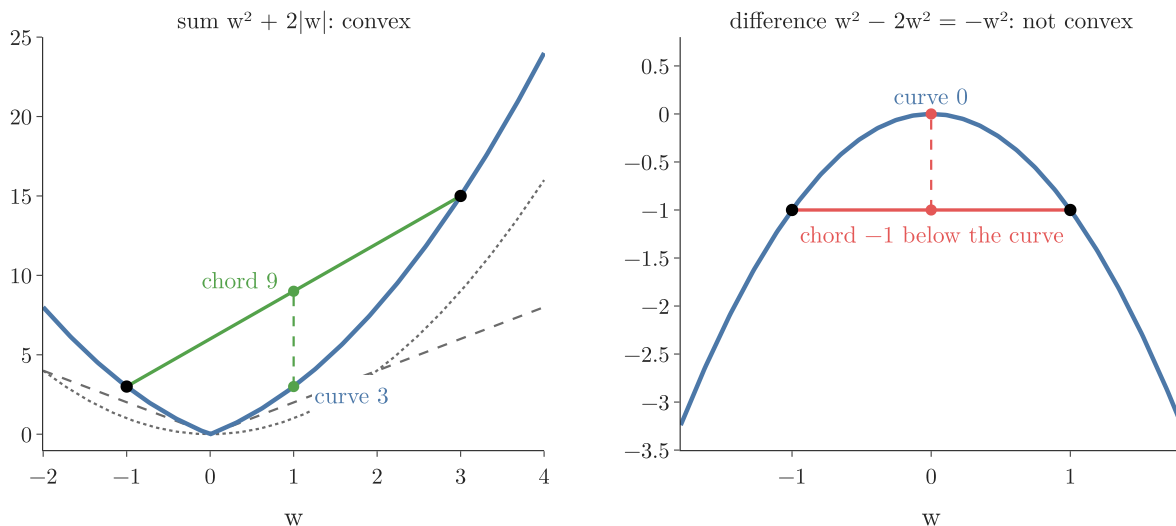


Figure 7: Left: the sum $w^2 + 2|w|$ (blue) of two convex pieces (grey); its chord from -1 to 3 stays above the curve, 9 against 3 at the midpoint. Right: the difference $w^2 - 2w^2 = -w^2$; its chord sits below the curve

Figure 7 puts the rule and its failure side by side. Watch where the chord sits: above the curve for the sum, below it for the difference.

This rule covers the regularised losses of earlier Notes:

- **Ridge:** squared error plus $\alpha \|\mathbf{w}\|^2$ (see the [Ridge regression maths Note](#)): convex plus convex.
- **Lasso:** squared error plus $\alpha \sum |w_i|$ (see the [Lasso regression Note](#)): convex plus convex, even though $|w|$ has a corner.
- **Regularised logistic regression:** log loss plus a penalty: convex plus convex.

What does not keep convexity. A difference or a product of convex functions can fail:

- **Difference:** w^2 and $2w^2$ are convex, but $w^2 - 2w^2 = -w^2$ is an upside-down bowl.
- **Product:** w^2 and $(w - 1)^2$ are convex, but their product $w^2(w - 1)^2$ has two dips (Figure 4 and Figure 5, right). Its chord from 0 to 1 is at height 0 , while the curve at 0.5 is 0.0625 .

Extra: One more rule: the pointwise **maximum** of convex functions is convex (Boyd and Vandenberghe §3.2.3). The hinge loss $\max(0, 1 - z)$ of the [SVM soft margin Note](#) is the maximum of two straight lines, so it is convex, and with the convex penalty $\frac{1}{2}\|\mathbf{w}\|^2$ the soft-margin SVM is a convex problem.

6. Convex optimisation problems

Key point: Minimising a convex function subject to convex inequality constraints and linear equality constraints is a convex optimisation problem: every local minimum is global, and strong duality holds.

6.1 The definition

Key point: The objective and every inequality function must be convex, and every equality constraint must be linear (affine).

1. **In words:** a convex objective, minimised over a feasible region built from convex pieces.
2. **Formula:** the problem

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0$$

is a **convex optimisation problem** (G-478) when f and every g_i are convex functions and every h_j is affine, $h_j(\mathbf{x}) = \mathbf{a}_j^\top \mathbf{x} - b_j$. Then the feasible region is a convex set (Section 2.2).

3. **Example:** the problem of the [Lagrange multipliers Note](#): minimise $x^2 + 2y^2$ (Hessian with eigenvalues 2 and 4, convex) subject to $3 - x - y \leq 0$ (linear, so convex). The problem is therefore a convex optimisation problem.

6.2 What convexity guarantees

Key point: No local-minimum traps, a zero-gradient (or KKT) point is the answer, and the dual reaches the same value as the primal.

For a convex optimisation problem:

- **Every local minimum is a global minimum.** The argument of the [convex and non-convex cost functions Note](#) (Section 3.2) works inside a convex feasible region too, because the segment towards a lower point stays feasible.
- **The first-order conditions are enough.** Without constraints, a point with zero gradient is the answer (Section 4.1). With constraints, a point that satisfies the KKT conditions of the [Lagrange multipliers Note](#) is the answer.
- **Strong duality.** The maximum of the dual function equals the primal minimum. In the example above, $D(4) = 6$ equals the primal minimum 6.

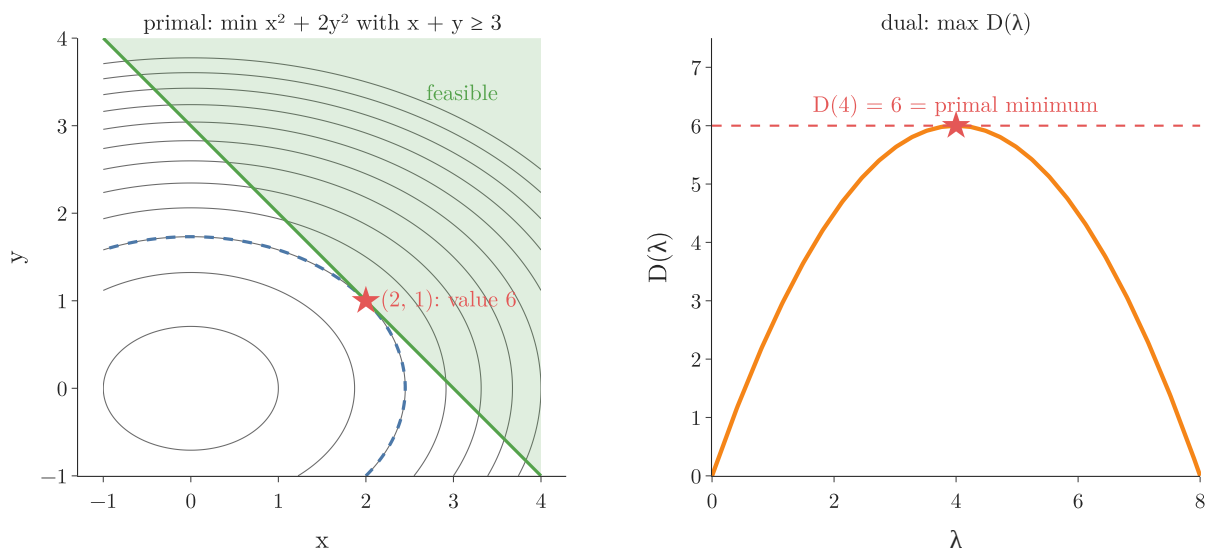


Figure 8: Left: contours of $x^2 + 2y^2$ and the feasible half-plane $x + y \geq 3$ (green); the lowest feasible point is $(2, 1)$, on the contour of value 6 (dashed). Right: the dual function $D(\lambda) = 3\lambda - 3\lambda^2/8$ peaks at $\lambda = 4$ with the same value 6

Figure 8 shows the example from both sides. Watch the two stars: the lowest feasible point on the left and the top of the dual on the right sit at the same height, 6.

Extra: Strong duality for a convex problem needs one mild extra condition, **Slater's condition** (G-1821): at least one point satisfies every inequality constraint strictly. For $3 - x - y \leq 0$, the point $(3, 3)$ gives $-3 < 0$, so it holds (Boyd and Vandenberghe §5.2.3).

6.3 Which ML problems are convex

Key point: The classical linear models are convex problems; trees, clustering and neural networks are not.

Problem	Convex?	Why
Linear regression, squared error	yes	Hessian $2X^T X$ is positive semi-definite
Ridge, Lasso, Elastic Net	yes	convex loss plus convex penalty
Logistic regression	yes	log loss is softplus of a linear score
SVM (hard and soft margin)	yes	convex objective, linear constraints (a quadratic program)
K-means	no	the result depends on the starting centroids

Problem	Convex?	Why
Neural networks	no	many minima and saddle points (Goodfellow et al. §8.2)

For the convex ones, the answer does not depend on the starting point or the solver, only on the data and the hyperparameters. The two best-known families of convex problems, linear and quadratic programs, are the topic of the [linear and quadratic programming Note](#).

7. Summary

Idea	Test	Example
Convex set	every segment between two points stays inside	disc yes, ring no
Convex function	chord on or above the graph	w^2 yes, $w^2(w-1)^2$ no
Rising slope	$f'' \geq 0$ everywhere (one variable)	$q''(0.5) = -1$: not convex
First-order test	tangent on or below the graph	softplus: $2.13 \geq 1.69$
Second-order test	Hessian positive semi-definite everywhere	eigenvalues 1, 3 yes; 5, -1 no
Building rules	non-negative sums and maximums keep convexity	Ridge, Lasso, hinge loss
Convex problem	convex f and g_i , affine h_j	Lagrange Note example

- Overlaps of convex sets are convex, so convex constraints give a convex feasible region.
- For a convex function, zero gradient means global minimum.
- Convex problems have no local-minimum traps and satisfy strong duality.

8. Sources

Built from

- CampusX, “Difference between convex & non-convex cost function; what happens when cost function is non-convex?”, YouTube, <https://www.youtube.com/watch?v=TXVtbgaEyms>
- Khan Academy (Khan, S.), “Concavity introduction”, YouTube, <https://www.youtube.com/watch?v=I>
- Khan Academy (Khan, S.), “Second derivative test”, YouTube, <https://www.youtube.com/watch?v=cW5hCsc9Yc>

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 7.3 (MML): convex sets, the building rules and convex optimisation problems, which no video in the list covers.
- Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Sections 3.1–3.2 (convex functions, the first- and second-order conditions, and the operations that keep convexity) and 5.2.3 (Slater’s condition).
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*. MIT Press. Section 8.2, challenges in neural network optimisation.

9. Key terms

Term	Meaning
Convex set	A set that contains the whole segment between any two of its points
Concave function	The negative of a convex function; every chord lies on or below its graph
Epigraph	The region on and above a function's graph; convex exactly when the function is convex
Jensen's inequality	For a convex function, the function of a weighted average is at most the weighted average of the function
Softplus	The function $\ln(1 + e^z)$, a smooth convex curve whose derivative is the sigmoid
Strictly convex function	A function whose curve lies strictly below every chord between two different points; it has at most one minimum
Concave up, concave down (G-2256)	Calculus names for a cup-shaped (convex) and a hump-shaped (concave) part of a curve
Second derivative test (G-2257)	At a flat point, $f'' > 0$ means a local minimum, $f'' < 0$ a local maximum, $f'' = 0$ no conclusion
First-order condition	A differentiable function is convex exactly when every tangent plane lies on or below its graph
Second-order condition	A twice-differentiable function is convex exactly when its Hessian is positive semi-definite everywhere
Convex optimisation problem	Minimising a convex function subject to convex inequality constraints and affine equality constraints
Slater's condition	Some point satisfies every inequality constraint strictly; together with convexity it guarantees strong duality